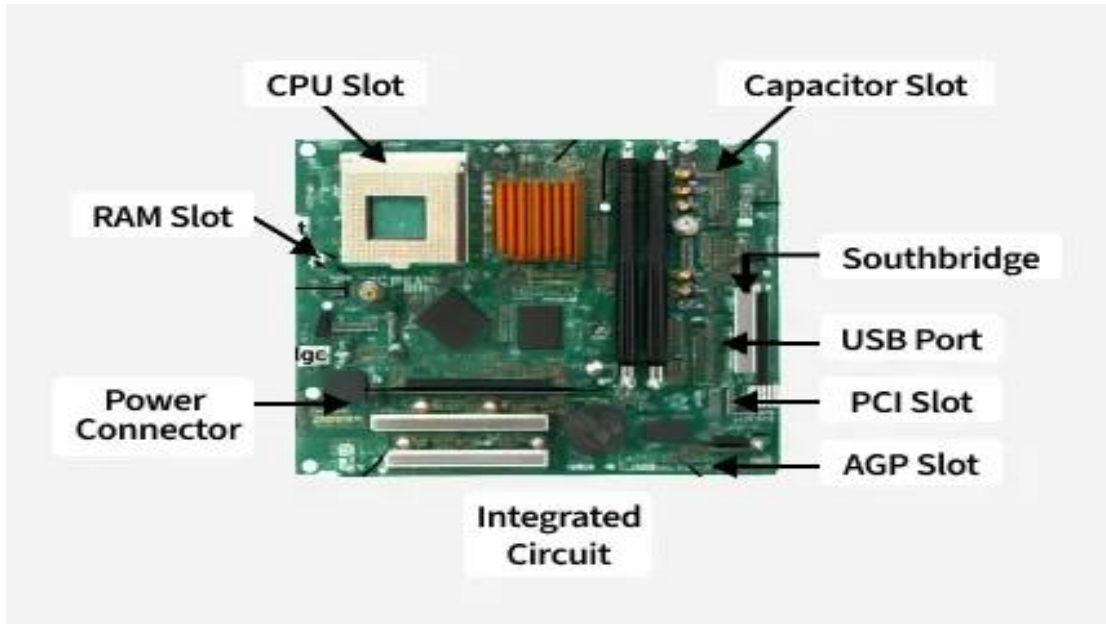


1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, Power connector, etc.)



2. Research and list 3 popular CPU Socket types (for example: LGA1151, AM4, LGA1200) and Name one compatible CPU model for each.

I. LGA1151

- LGA1151 is a Intel CPU socket. It support several Intel processor generations.
- Compatible CPU Model : Intel core i7-7700k
- LGA1151 Intel 6th-9th generation processors depending on motherboard chipset and BIOS version.

II. AM4

- AM4 is an AMD socket. It supports long platform longevity.
- Compatible CPU Model : AMD Ryzen 5 5600X
- It supports Ryzen processors from Ryzen 1000s to 5000 series.

III. LGA1200

- LGA1200 is a Intel CPU socket. It is mainly designed for desktop processors.
- Compatible CPU Model : Intel core i5-10400
- LGA1200 supports Intel 10th and 11th generation desktop processors.

3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify

- **How It works Together**

1. Request :

- The CPU processes the mouse click and determines that Spotify should be launched.

2. Loading The App :

- The CPU requests the Spotify program files from the SSD or hard drive. Then, chipset helps manage communication between the storage device and the rest of the system. Later on, all files are transferred into RAM.
- The CPU executes instructions and network hardware communicates through motherboard connection
- After that, Data from Spotify's servers sent to the computer.

3. Music Playing :

- The CPU handles tasks such as decoding compressed audio data. Then, audio device converts data into sound which later on reaches through speakers or headphones.
4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labelling each one.

